



work importance locator

User's Guide



U.S. Department of Labor
Employment and Training Administration





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Users intending to develop other products, software or systems applications using O*NET Career Exploration Tools products must contact the National O*NET Consortium at <http://www.onetcenter.org> or National O*NET Consortium, North Carolina Employment Security Commission, P.O. Box 27625, Raleigh, NC 27611, for the Developer's Agreement.

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The development of the **O*NET Work Importance Locator** was completed under the direction of David Rivkin and Phil Lewis. The Human Resources Research Organization (HumRRO) was the primary contractor in developing the instrument. Their team was led by Dr. Rodney McCloy. Dr. Gary Carter, John Boyle, and Dan Connell initiated and participated in many of the development research efforts.

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Numerous research phases were conducted in developing the **O*NET Work Importance Locator**. Each phase required multiple steps and the participation of a wide variety of staff, contractors, and pilot sites. Grateful acknowledgment for their prominent roles in this process is made to Jeannie Stroupe, Helen Tannenbaum, Ray Dobrin, Enid Israelson, Ann Kump, Diana Williams, Dr. Stephen Atkins, Ronald Boese, Brenda Dunn, and Jerry Pickett.

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OVERVIEW

The U.S. Department of Labor's (USDOL's) Office of Policy and Research has developed **O*NET™ Career Exploration Tools**. The tools are designed to assist a wide variety of users in identifying information about themselves. They can use this self-knowledge to guide their exploration of occupations included in **O*NET OnLine**, the automated replacement for the *Dictionary of Occupational Titles* (U.S. Department of Labor, 1991).

The tools stress self-directed whole-person assessment. Users are able to take a variety of valid and reliable assessment tools, each providing important information that can help them explore the world of work. **O*NET Career Exploration Tools** include:

- The **O*NET Work Importance Locator**, which helps users identify what is important to them in a job. It can help individuals identify occupations they may find satisfying. The questions are answered and scored in a booklet.
- The **O*NET Work Importance Profiler**, which is similar to the **O*NET Work Importance Locator**, except the questions are answered on a computer.
- The **O*NET Interest Profiler**, which helps individuals identify their work-related interests. It is a paper-and-pencil instrument.
- The **O*NET Computerized Interest Profiler**, which is similar to the **O*NET Interest Profiler**, but is a computerized instrument.
- The **O*NET Ability Profiler**, which helps individuals identify what they can do well.

Thus, these assessment tools help individuals discover three important pieces of information that are valuable when exploring careers:

- 1) what they like to do,
- 2) what is important to them in the world of work, and
- 3) what they do well.

O*NET Career Exploration Tools are designed to be interactive and flexible. Individuals are able to take one or all of the instruments; or they can use them with other career exploration instruments developed by private organizations, depending on their particular needs. Additionally, the **Interest** and **Work Importance** tools can be self-administered, with no outside assistance. All of the tools present individual users with results presented on score reports that are self-interpretable and are linked to occupations in **O*NET OnLine**. The other career exploration tools and their associated reports also are useful for group settings, such as vocational training programs, classrooms, or out-placement programs.

This User's Guide was developed for the **O*NET Work Importance Locator (WIL)** in order to help users:

- 1) learn how to administer the WIL,
- 2) interpret WIL results using the **O*NET Work Importance Locator Score Report**, and
- 3) understand how the WIL was developed.

User guides are available also for the other **O*NET Career Exploration Tools**. For information on these guides or other O*NET products contact the Center for O*NET Development:

Internet:
<http://www.onetcenter.org/>

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National Center for O*NET Development
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Raleigh, NC 27611

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SPECIAL NOTICE: PROPER USE OF O*NET WORK IMPORTANCE LOCATOR RESULTS

Please pay particular attention to the proper use of **O*NET Work Importance Locator** results. Part of your responsibility as an administrator/user of the **O*NET Work Importance Locator** is to ensure its proper use.

O*NET Work Importance Locator results **should be used** for career exploration and vocational counseling purposes only. Results are designed to assist clients in identifying their work values (i.e., what is important to them in an occupation) and in using them to identify occupations that may satisfy their work values.

O*NET Work Importance Locator results **should not be used** for employment or hiring decisions. Employers, educational programs, or other job-related programs should not use **O*NET Work Importance Locator** results in applicant screening for jobs or training programs. The relationship between results on the **O*NET Work Importance Locator** and success in particular jobs or training programs has not been determined.

Please read the **Special Notice: User Agreement** on page i of this Guide before administering the **O*NET Work Importance Locator**.

For further information on proper use of the **O*NET Work Importance Locator** results, contact:

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INTRODUCTION

One of the U.S. Department of Labor's (DOL) **O*NET™ Career Exploration Tools** is the **O*NET Work Importance Locator** (WIL). This is a new work values assessment instrument. Work values are the aspects or conditions of work that are important to people in a job or career. The WIL helps clients accurately and reliably identify their highest work values. Clients can use this information to:

- 1) learn a valuable piece of self-knowledge and career awareness, and
- 2) directly link to the entire world of work via the 900+ occupations within **O*NET OnLine**.¹

The **O*NET Work Importance Locator** was developed using the most up-to-date knowledge of vocational theory and practice. The instrument is based on a previously developed measure of work values, the Minnesota Importance Questionnaire (MIQ; Rounds, Henly, Dawis, Lofquist, & Weiss, 1981). Clients use a simple card-sorting format to rank the importance of 20 cards, each describing an aspect of work that satisfies one of six broad work values. The six values are updated versions of the work values defined in Dawis and Lofquist's (1984) Theory of Work Adjustment: Achievement, Independence, Recognition, Relationships, Support, and Working Conditions. The Theory of Work Adjustment (TWA), which is grounded in a rich and extensive research history, is a comprehensive model of vocational adjustment based on the concept of correspondence between individual and environment. Importantly, work adjustment has been linked to the satisfaction of workers, as well as how well workers perform on their job.

This Guide was developed for programs (e.g., schools, employment service offices, career information and delivery systems, out-placement programs) that will incorporate the WIL into their career exploration services. The Guide can help users understand how to properly incorporate the WIL into their programs by providing technical development information, administration instructions, and guidelines for interpreting results.

¹ The occupations within **O*NET OnLine** are based on the O*NET/SOC classification system. The Office of Management and Budget has mandated the use of the SOC system for government agencies. Thus, the O*NET classification system will be compatible with other sources of occupational and labor market information. Currently, 900 of the 974 occupations have O*NET data and are included within the **O*NET Work Importance Locator** materials.

This User's Guide is divided into the following three sections:

- 1) **Administering the O*NET Work Importance Locator**—provides a description of individual, group, and combination administration options. This section also walks the user through the different parts of the WIL.
- 2) **Interpretation of O*NET Work Importance Locator Results**—provides:
 - 1) an overview of Dawis and Lofquist's (1984) Theory of Work Adjustment, the underlying structure behind the WIL;
 - 2) a description of how the WIL score report can be used to interpret WIL results;
 - 3) explanations of how the score report helps users identify occupations to explore further; and
 - 4) other activities that can be used to help users better understand and use their WIL results.
- 3) **Development of the O*NET Work Importance Locator**—provides an overview of the procedures followed to develop the **O*NET Work Importance Locator**.
Note: This section provides an overview of fundamental technical information that is important to vocational researchers. It also contains information that will provide educators and vocational counselors with a better understanding of the steps that went into ensuring that the WIL meets stringent technical standards. For a more detailed description of the development of the WIL, see ***Development of the O*NET Work Importance Locator*** (McCloy, Waugh, Medsker, Wall, Rivkin, & Lewis, 1999b).

Together, these three sections, along with the WIL and its associated score report, should provide users with the information they need to use the instrument effectively in their organization's career counseling program.

ADMINISTERING THE O*NET WORK IMPORTANCE LOCATOR

There are several ways to administer the **O*NET Work Importance Locator** (WIL). Although the instrument was designed primarily to be self-administered—with clients taking the instrument on their own with no outside assistance—the instrument also can be administered in a group setting. Additionally, combinations of administration methods can be used. You can have clients complete part of the WIL on their own and part of it in a group. They can take most of it by themselves and then receive some outside assistance from a career counseling professional.

BEFORE ADMINISTERING THE O*NET WORK IMPORTANCE LOCATOR

Each of the different administration methods is described below. Before administering the WIL, it is important to review all the methods. You will notice that the different methods build on each other, and that the information presented for one method can help accomplish another method. It is especially important to read the Self-Administration subsection because it explains the different parts of the **O*NET Work Importance Locator**. After reading about each administration method, you will be able to use a method, or parts from each method, that best suits your particular situation.

Also, before administering the WIL, it is very important that you take the instrument yourself to become familiar with all the WIL parts and procedures.

SELF-ADMINISTRATION

The **O*NET Work Importance Locator** was designed to be self-administered. The instrument contains all the necessary instructions for individuals to complete the measure on their own. Below are some commonly asked questions that users of the WIL have regarding self-administration. The answers provided will enable you to make sure clients enjoy their WIL experience and receive accurate, meaningful results.

What reading level is the O*NET Work Importance Locator designed for?

The WIL was designed for use with a wide variety of populations, including workers in transition, unemployed workers, college students, and high school students. It is suggested that clients have a **minimum of an eighth grade reading level** to take the WIL. Clients below an eighth grade reading level may not be able to adequately understand the information in the WIL.

What age level is the O*NET Work Importance Locator designed for?

The WIL was designed for clients who are approximately **16 years of age and older**. The development of values is a process that occurs throughout a person's lifespan and is modified by experiences. However, as individuals mature, values tend to become more stable. By the time people enter the 10th grade (approximately 16), their values are stable enough to be measured reliably. While the development of values takes place in family, social, cultural, educational, and work settings, individuals younger than 16 may not have had enough exposure to a work environment to allow them to understand the descriptions of work included in the WIL (e.g., *"...I would have supervisors who train their workers well."*).

Where is the best place to "self-administer" the WIL?

The WIL is flexible in terms of when and where individuals can take the instrument. Clients can take it at home, in school, in the library, or in another **quiet location** where they will be able to concentrate and respond seriously to WIL questions. In addition, they will need to have access to a flat area (e.g., desk or table) to place their Work Value Card Sorting Sheet while they are working on the WIL.

Note: The Work Value Card Sorting Sheet, along with the other parts of the WIL, will be described in detail in the following subsection.

How long will it take clients to complete the WIL?

The WIL takes anywhere from **15 to 45 minutes to complete**. For most users, approximately three-fourths of their time is spent answering questions, and the remaining portion of their time is spent scoring the instrument. After completion, additional time is needed to read and use the WIL Score Report, which helps clients interpret their results and explore occupations presented in **O*NET OnLine**.

Note: The **Interpretation of O*NET WIL Results** section of this Guide provides more details regarding how the Score Report helps clients understand and use their results for career exploration.

Can clients complete the O*NET Work Importance Locator in one session?

Yes. The instrument was **designed to be completed in one session**. Clients should answer all questions and score the instrument at one time. This will provide continuity to the administration process and will help clients better understand the information the WIL is trying to convey.

Can clients complete the O*NET Work Importance Locator in two sessions?

It is **not recommended**. Clients should be strongly encouraged to complete their WIL in one sitting. The WIL contains loose cards that must be placed and moved about on the Work Value Card Sorting Sheet. Storing clients' cards and sorting sheets between sessions increases the likelihood that their cards will be misplaced—shuffled—leading to inaccurate results. In addition, research has demonstrated that a person's first response is often most accurate. Clients may be tempted to change some of their responses during the second session, leading to ambiguous results.

Note: If clients are not sure about their results after completing the WIL, they are encouraged to take the entire instrument over again at a later date (see the **Interpretation of O*NET WIL Results** section of this Guide for more detail).

Is a calculator needed for clients to score their WIL?

No. Research has demonstrated that **clients can accurately score the WIL without the aid of a calculator**. However, basic multiplication and addition is required to score the WIL. If you feel your clients may have difficulty with either of these tasks, a calculator is recommended.

What are the 20 cards for?

Each card contains a “need” statement that is related to one of the six work values measured by the WIL. When scoring the WIL, users will use these cards to calculate their work values scores.

PROCEDURES FOR SELF-ADMINISTRATION OF THE O*NET WORK IMPORTANCE LOCATOR

The WIL contains all the necessary information for clients to complete the instrument on their own. The following is an overview of the instructions individuals read and follow to complete WIL items and to score the WIL.

Instructions for Completing the O*NET Work Importance Locator

To start their WIL, clients simply open the instrument and begin reading.

Overview

First, they are presented with an overview of the **O*NET Career Exploration Tools**. This section provides users with a brief description of the different O*NET assessment tools. It lets the clients know that the assessment tools can help them decide what is important to them in a job, what work they probably will be able to do well, and what type of work they would like to do. It also indicates to clients that they can use different pieces of information about themselves to explore careers.

Getting Started

Next, clients complete the **Getting Started** section on page 1 of the WIL. Here clients write their name and the date in the spaces provided. They also read **Welcome to the O*NET Work Importance Locator**, which lets them know that they will be ranking different aspects of work in order to learn about their work values and help them decide what is important to them in a job. It informs them that by completing the WIL, they will have six work value scores that will show them how important each value is to them. They will be able to use their results to select occupations to explore that may lead to satisfying jobs and careers.

This section also stresses the proper use of **O*NET Work Importance Locator** results. It lets clients know that results are for career exploration and vocational counseling purposes only, and that they should not be used for employment selection or screening decisions. It also informs clients to contact the National Center for O*NET Development for further information on the proper use of their **O*NET Work Importance Locator** results.

The **Before You Begin** section informs clients that there are no right or wrong answers and encourages them to answer their WIL honestly and carefully. It also includes the following list to help clients make sure that their package has all the materials necessary for them to complete the WIL:

- twenty **Work Values Cards**, each containing a need statement related to one of the six work values,
- a **Work Value Card Sorting Sheet**,
- an **O*NET Work Importance Locator Score Report**, containing lists of occupations, and
- the set of instructions they are currently reading.

Note: All of the materials listed above always should be present in each of your clients' WIL packets. If you find that a number of packages are missing materials, you should contact the Center for O*NET Development (contact information listed on page 2 of this Guide).

Completing the O*NET Work Importance Locator

Clients move on to **Instructions** on page 2. This section first provides clients with a brief overview: introducing them to the idea of using cards rather than answering questions, describing the cards (i.e., statements about different aspects of work), and summarizing the task (i.e., sorting the cards into groups based on how important the statement on each card is to the client on his/her ***ideal job***).

Note: The term "ideal job" is in bold and italics. This is the first place that this important concept is described, "the kind of job you would most like to have." Research has demonstrated that individuals' results may be inaccurate if they are

thinking about their present or last job while completing the WIL, rather than their ideal job.

Clients are then directed to descriptions of the specific steps they should follow to complete and score the WIL. They are instructed to follow the steps in order. The first two steps involve completing the instrument; the third describes how to score the instrument, and the fourth initiates clients' interpretation of their results.

Step 1. Read the Cards. This step involves clients reading the work aspect (i.e., need statement) described on each card and thinking about the importance of each aspect. Clients are instructed to:

- find the 20 Work Values Cards in their packet,
- read all 20 cards before they go on to Step 2, and
- think about how important it would be for them to have a job like the one described on each card.

Step 2. Sort the Cards. This step involves clients placing the Work Values Cards on the Sorting Sheet in an order that best reflects their importance to the client. Description of this step is included on the Work Value Card Sorting Sheet, as well as in the main body of the WIL, to give clients the maximum opportunity to read and refer to the information. Clients are instructed to:

- find their Work Value Card Sorting Sheet in their packet,
- look at the five columns printed under the Importance Scale at the top of sheet,
- notice that each column has a number 5 (Most Important) to 1 (Least Important),
- put each card in the column that best matches how important it is for them to have a job like the one described on each card, and
- put exactly four cards into each column.

Note: The statement, "Put exactly 4 cards in each column," is in bold and italics. Four cards must be placed into each column. There is no other placement option (e.g., five cards in one column, three in another). Occasionally, clients may complain about having to choose between statements of equal importance to them when placing cards in columns. When this occurs, mention to clients that no single job provides a person with everything they would like, and that they should not spend a great deal of time "over-thinking" the placement of the cards—instead they should use their first impression.

Additional description of the scale is provided to help clarify the use of the Importance Scale. Clients are informed that when they have completed the sorting task, the four most important statements should be in Column 5, and the four least important should be in Column 1. In addition, clients learn that they will have to move

the cards around until each column has exactly four cards. They are directed to look at the example on the next page of the WIL for more help.

For Example. This section of the WIL provides clients with a visual example of performing the card sort. Two sample scenarios for the placement of Card A are visually presented:

- First, if Card A describes something more important to the client than what is described on any of the other cards, clients are directed to put the card in Column 5. Card A is placed in Column 5 on the diagram of the sample Sorting Sheet.
- Next, if Card A describes something less important when compared to all the other cards, clients are directed to put the card in Column 1. Card A is placed in Column 1 on the second diagram of the sample Sorting Sheet.

The example also notes that if Card A was neither the most important nor the least important, then it would be placed in one of the other columns (4,3,2) that best matches how the client feels.

The example concludes by reminding clients to “do the same thing for all of the cards.” In addition to providing a visual learning aide, the example also is intended to provide clients with another set of near-complete directions in the event that they skipped the written directions on the prior page.

Scoring the O*NET Work Importance Locator

After clients have successfully sorted all 20 cards, they are directed to move on to Step 3 located on page 4 of the WIL.

Step 3. Figure Out Your Work Values Scores. This section starts off by listing the six work values:

Achievement	Relationships
Independence	Support
Recognition	Working Conditions

Clients then are directed to use the Work Value Worksheet on the opposite page of the WIL to figure out their scores for the six work values. Clients are instructed to:

- look at their Work Value Card Sorting Sheet;
- score each card by looking at the number of the column they put it in (the cards in Column 5 each receive a score of 5, the cards in column 4 each receive a 4, and so on);

(Note: The critical concept for clients to understand is that the column number represents the score. The word “score”, along with the column number, is bolded in the directions to help clients grasp this very important link.)

- do the same thing for all other cards;
- for each box on their Work Value Worksheet, add up the numbers in the Score Column and write the TOTAL in the space provided; and
- then multiply each TOTAL score by the number shown below that box, and to write their score next to the “=” sign in the box provided.

Clients are given a **NOTICE** at the bottom of page 4 stating that they do not need to multiply the total score to get their Working Conditions value score. However, it is extremely important for clients to multiply the total scores for the other five work values. Clients who fail to do this step will **not** have accurate, functional scores.

For Example: A box with an example is presented along with the instructions listed above. The example directs clients to look at the Achievement box on the next page and to see that Card A is listed first. Clients are then directed to find the column on their Work Value Card Sorting Sheet where they put Card A and to write the number of that column in the scoring space next to Card A in the Achievement box. The example is intended to provide those clients who elect not to read the instructions with the information they need to score the instrument.

Interpreting the O*NET Work Importance Locator

After clients have successfully scored the WIL and have all six work value scores, they are directed to turn to page 6, where they complete Step 4 of the WIL. After they are finished with this step, clients are directed to their WIL score report, where they begin to interpret their WIL results.

Step 4. Copy Your Work Values Scores. This step directs clients to find their two highest work values scores on page 5 and to write the names of those work values, along with the scores, in a space provided at the top of page 6.

Step 5. Go to Your O*NET Work Importance Locator Score Report. This step directs clients to their WIL Score Report, where they can learn what their scores mean and how to use this information to explore careers.

Note: The WIL focuses clients on their two highest work values. This is the reason for the WIL being called a “locator”—because it is designed to help clients quickly discover or locate their top work values. Those clients interested in using information related to all six work values when exploring careers should take the computerized **O*NET Work Importance Profiler**, which allows clients to use all six of their work values scores to explore careers.

Helpful Reminders

The **O*NET Work Importance Locator** was extensively pilot-tested by individuals from a variety of age, experience, and education levels. The vast majority of individuals did not have trouble completing and scoring the WIL on their own. However, there are a few reminders you can give clients to help ensure that they get the most out of the instrument and that they get a true picture of their work values.

- A. **Remind clients to read all instructions carefully.** All the necessary information needed to successfully complete the WIL is included in the instrument itself. However, sometimes clients may skip over important information. They may think they don't need to read every word because they "get" how to complete the WIL. The problem with skipping instructions is that users might miss something important, which will influence how they complete and score the instrument. If this happens, their results might not reflect their true work values.
- B. **Remind clients to read all 20 cards before going to the sorting task in Step 2.** Sometimes clients will start sorting the cards before reading through all of them. In order for clients to get the most meaningful results, they should read each of the cards before sorting them, so that they have a good understanding of the need statements.
- C. **Clients should be encouraged to read each statement, make a choice, and move quickly to the next statement.** It is likely that individuals' first reactions to each statement are the most indicative of their true feelings. "Over-thinking" the placement of the cards may lead to less accurate and less satisfying results.
- D. **Remind clients to think about their ideal job.** Clients are directed to think about their *ideal job*, "the kind of job you would most like to have," when considering the importance of each need statement. It is very important to remind clients to focus on their ideal job when completing the WIL, rather than their present or last job. Research has demonstrated that individuals' results may be inaccurate when they are using the wrong reference for rating the need statements.
- E. **Make sure clients put four cards in each column.** In order for the WIL's scoring system to work properly, clients must place four need statements—cards—into each column. There is no other placement option (e.g., five cards in one column, three in another). If clients complain about having to choose between statements of equal importance to them when placing cards in columns, mention to clients that no single job provides a person with everything they would like.
- F. **Remind clients to take their time, answer honestly, and enjoy themselves.** The WIL is not a test. There are no right or wrong answers, so clients should be encouraged to answer as honestly and frankly as possible. The WIL is an assessment tool designed to help clients identify information about themselves

that they can use to explore the world of work. Remind them that there are no time limits! Clients should relax and take advantage of the information the WIL has to offer.

G. Make sure clients transfer the column numbers to the Worksheet.

It is extremely important that clients transfer the correct column number (5, 4, 3, 2, or 1) next to the matching work value card letter on their Work Value Worksheet. Verify that clients are transferring the column numbers correctly. If errors are made in this procedure, clients' scores will be inaccurate and not valid.

H. Remind clients who are scoring the WIL to perform the multiplication step as directed. It is extremely important for clients to perform the multiplication step as directed for five of the work values scores. Clients who fail to do this step will not have accurate, functional scores.

GROUP ADMINISTRATION

The **O*NET Work Importance Locator** also is suited to group administration. The term "group administration" has a variety of definitions that is often dependent on the type of instrument being administered. For the WIL, group administration means that a professional (e.g., counselor, teacher, program leader) leads the administration of the WIL and provides assistance to WIL users. It can include, but is not limited to, the following procedures:

- Reading the instructions aloud to clients as they read along with the leader. You can also have individuals take turns reading parts of the instructions aloud.
- Answering clients' questions regarding WIL instructions. For example, clients may have a question concerning what they should think about when they are reviewing their cards (i.e., ideal job), how they should use their Work Value Card Sorting Sheet, or what they should do if they want to put more than four cards in a particular column.
- Monitoring clients as they complete WIL items, making sure that they put exactly four cards in each column, and also that they don't appear to be "over-thinking" or second guessing their card placements.
- Answering clients' questions about specific card statements. Clients, especially those with little work experience, may need help understanding some of the work-related concepts included in the need statements. For example, they may not be familiar with the concept of supervisors "backing up" their workers or having a job with good "working conditions."

- Working step-by-step through the scoring procedures. The leader may choose to read out loud the **Step 3** instructions located on page 4. The leader also could read the **For Example** section to the clients as they follow along. Another option includes having the clients first score their **Achievement** work value, verifying the achievement score they came up with, and then having them move on and score their remaining work values.

Below are some questions and answers regarding group administration that can help you decide if you want to administer the WIL in a group setting.

What locations are better for group administration?

Just like self-administration, a group administration should take place in locations where individuals will be able to concentrate and focus on the WIL. A classroom or library is ideal for a group administration. In addition, clients will need to have access to a flat area (e.g., desk or table) on which to place their Work Value Card Sorting Sheet while they are working on the WIL. Clients can sit at individual desks, or they can sit around a table to facilitate group interaction during discussion of results (see the **Interpretation of O*NET WIL Results** section of this Guide). It is very important to remember, however, that clients should sort their WIL cards on their own, based on what *they think* is important on their ideal job. They should not be influenced or persuaded by the ways that fellow group members decide to sort their own cards.

When is it appropriate to administer the WIL in a group setting?

The WIL is appropriate for group settings that occur in a variety of programs (e.g., schools, employment service offices, outplacement centers). For example, the WIL could be part of a vocational training program in which clients talk about their results and the ways that the WIL helped them discover important information about themselves. The leader also could have clients discuss where they learned about their perceptions of work (e.g., experience on current or previous jobs; parents and friends; television, books, and other media) and how these perceptions may have played a role in their results.

A program may have certain clients who require “extra help” to complete the WIL. For example, some clients may have less than an eighth grade reading level, they may be very unfamiliar or nervous about taking assessment tools, or they may have trouble focusing on a task. Group administration is also appropriate for individuals who might have some trouble with the English language (e.g., English-as-a-Second-Language Students).

How long will it take to conduct a group administration of the O*NET Work Importance Locator?

Group administrations should take **approximately 45 minutes**. This includes leading clients through taking the items and scoring the instrument. Interpretation of results will take an additional period of time that varies, depending on the amount of detail

and depth the leader elects to cover (see **Interpretation of O*NET WIL Results** section for a discussion of some options).

Can a group administration be held over two sessions?

No, it is **not recommended**. Similar to self-administration of the WIL, leaders are strongly encouraged to have their clients complete and score the WIL in one sitting. The WIL contains loose cards that must be placed and moved about on the Work Value Card Sorting Sheet. Storing clients' cards and Sorting Sheet between sessions increases the likelihood that their cards will be misplaced—shuffled—leading to inaccurate results. In addition, research has demonstrated that a person's first response is often the most accurate. Clients may be tempted to change some of their responses during the second session, leading to ambiguous results. Interpretation of results, discussed in the next section of this Guide, can take place in a second session.

Are there certain things I should pay particular attention to during a group administration?

Yes. The same “reminders” that were presented in the self-administration subsection apply to group administration. These reminders are summarized below:

- A. Remind clients to read all instructions carefully.
- B. Remind clients to read all 20 cards before going to the sorting task in Step 2.
- C. Encourage clients to read each statement, make a choice, and move quickly to the next statement.
- D. Remind clients to think about their *ideal job*.
- E. Make sure clients put four cards in each column.
- F. Remind clients to take their time, answer honestly, and enjoy themselves.
- G. Make sure clients transfer the correct column numbers to the Worksheet.
- H. Remind clients who are scoring their WIL to perform the multiplication step as directed.

How many clients can participate in a WIL group administration at one time?

The maximum number for a group administration should be about 25. Any group larger than this will make monitoring the completion and scoring of the WIL difficult. There is no minimum size requirement for group administration of the WIL. For example, you might decide to do a group administration for only two clients, if you have the time or if the clients require special attention.

COMBINING ADMINISTRATION METHODS

Another option for programs using the WIL is to combine methods of administration. That is, some assistance could be provided and/or clients might take the majority of the instrument on their own and only receive assistance during certain portions of the administration. Two examples of combination strategies are described below.

Providing Assistance to an Individual Client

A program might have clients who have poor reading skills and/or trouble focusing on tasks. For this type of client, a one-on-one session might be appropriate, with the counselor and client taking turns reading the instructions. The client could receive help sorting the first couple of cards. After finishing the WIL, the client could receive help scoring the instrument, perhaps scoring the first work value together, and then working on the rest of the values on his or her own. The counselor could check the individual's work as progress is made on scoring the additional values.

Note: If, after receiving help on the first couple of items, the client cannot complete the remaining items on his or her own, the instrument is probably not appropriate for the individual. It is important to remember that the instrument has approximately an eighth grade reading level.

Of course, the amount of assistance provided to clients depends on their skills and abilities. Thus, unlike the type of client described above, a program might have a client who can read instructions on his/her own and complete the items without assistance, but who needs help scoring the instrument. With this client, simply start the person on the instrument, give him or her the proper "reminders" (see **Self-Administration** subsection for a list of reminders), and then assist the client in scoring the instrument.

Combining Self-Administration and Group Administration Methods

The **O*NET Work Importance Locator** is designed to be flexible, so that different methods of administration can be used or combined. It is perfectly acceptable to combine self- and group administration methods. For example, one option includes reading the instructions together, then having the clients complete the instrument and scoring at home. Or, clients might be directed to read the instructions ahead of time on their own and then actually complete the instrument in a group setting, where they can get assistance if needed.

Again, the level of assistance provided to a single client or a group of clients depends on their capabilities. Based on administrator/counselors' experience with the individuals, a decision can be made regarding what clients can do by themselves and where they might need some help.

In the section that follows, **Interpretation of O*NET WIL Results**, the flexibility of the **O*NET Work Importance Locator** is demonstrated once again. As with

administration, the score interpretation materials were designed for clients to use on their own. However, there may be a need to provide some additional assistance to clients, and a group interpretation session is an optional activity.

INTERPRETATION OF O*NET WORK IMPORTANCE LOCATOR RESULTS

The **O*NET Work Importance Locator** was designed to enable the majority of people to interpret and use their results for career exploration on their own. Along with the instrument, clients receive an **O*NET Work Importance Locator Score Report**. This report helps them understand what their results mean and provides instructions for applying their results to start exploring careers using **O*NET OnLine**, an application available on the Internet at <http://www.onetcenter.org>. **O*NET OnLine** helps clients explore the occupational information in the O*NET database.

Information presented in the following subsections of this Guide includes:

- A brief overview of the Theory of Work Adjustment (Dawis & Lofquist, 1984), the theory behind the **O*NET Work Importance Locator**. Familiarity with the theory will provide an understanding of the use of work values for career exploration, and help clarify the design and intent of the WIL and its associated score report.
- A description of each part of the Score Report. The descriptions include an explanation of the information each part communicates to clients.
- Possible challenges that clients might face with a particular portion of the Score Report. Descriptions of the challenges are presented along with solutions that you can use to help clients better understand the Score Report.

Note: As mentioned previously, the **O*NET Work Importance Score Report** was designed for self-use. You, however, may wish to work with clients in interpreting their results.

- Suggestions for program activities that can help clients interpret and use their results.

THE THEORY OF WORK ADJUSTMENT

Overview

The **O*NET Work Importance Locator** is based on Dawis and Lofquist's (1984) Theory of Work Adjustment. The theory, which has evolved over four decades of research, provides a comprehensive model conceptualizing the interaction between individuals and work environments.

The interaction is made up of an initial "fit" between individuals and their environment, as well as dynamic elements that characterize ongoing adjustments made by both the individuals and the work environment. In other

words, individuals with particular characteristics are best suited for jobs that have work demands that correspond with those characteristics. Individuals depend on the work environment to reinforce their characteristics or “needs,” and the work environment depends on individuals to meet the demands or “requirements” of the job. The greater the correspondence between the individual and the work, the greater the likelihood of job satisfaction, performance, and tenure. Once an individual is in a particular job, however, over a period of time the job will affect the characteristics of the worker, and the worker will affect the demands of the job.

While the theory is quite extensive and complex, there are four major concepts that are critical to its understanding. A summary of each concept is listed below:

- The fit between the needs of an individual and the reinforcers provided by the work environment affects how satisfied the individual is with work.
- The fit between the abilities of an individual and the ability requirements of the occupation affects what the theory calls “satisfactoriness.” This term refers to how satisfied the work is with the individual. To make the theory a little more user-friendly, “satisfactoriness” can be thought of as how well the individual “performs” on the job. **Note:** This Guide will use the term “performance.”
- Performance influences satisfaction, and satisfaction influences performance.
- How long an individual stays on the job (i.e., tenure) is affected by both satisfaction and performance.

Stated simply, the four concepts above advocate that when exploring careers, individuals are: more likely to be satisfied by jobs that meet their needs, are more likely to perform better if they have the abilities necessary to do the job, will perform better if satisfied, and will be more satisfied the better they perform. Lastly, individuals will stay on the job longer if satisfied and performing well.

How do work values tie into the Theory of Work Adjustment?

Work values have a critical role in the practical application of the theory. Needs, which were referred to above, are specific aspects of work that an individual requires to be satisfied. They are specific work characteristics that are important to individuals. Examples include: security, variety, responsibility, and creativity. However, individuals, especially those exploring careers, may not think about what is important to them in the world of work in such specific terms. Instead, they tend to think about work more globally. They often have general standards of what is important. This is where work values enter into the equation. The vast number of specific needs identified by years of empirical research can be grouped together according to broad themes of importance. These groupings make up what the lay person generally recognizes as work values. Examples of work values include: achievement, recognition, and independence.

Therefore, a more practical application of the theory involves replacing the concepts of an individuals' needs with work values. For example, it is the fit between the work values of an individual and the reinforcers provided by the work environment that affects how satisfied the individual is with work. If work gives individuals what they value, then they tend to be satisfied with their jobs.



When they are dissatisfied, it is most likely because their important work values are not being met. It is necessary to stress, however, that people differ in what they consider important on their ideal jobs.

Work Values in the O*NET Work Importance Locator

The **O*NET Work Importance Locator** directs individuals to sort 20 statements describing specific needs into five levels of importance. Each of the needs matches up to one of the six work values the instrument was designed to measure: Achievement, Working Conditions, Recognition, Relationships, Support, and Independence (see page 24 of this Guide for a summary of each work value). The rank ordering of an individual's needs provides the information necessary to determine the person's most important work values. Listed below are the six work values along with the specific needs that fall under each value (needs are italicized):

- Achievement
 - Ability Utilization*
 - Achievement*
- Independence
 - Creativity*
 - Responsibility*
 - Autonomy*
- Recognition
 - Advancement*
 - Recognition*
 - Authority*
- Relationships
 - Co-Workers*
 - Social Service*
 - Moral Values*
- Support
 - Company Policies and Practices*
 - Supervision, Human Relations*
 - Supervision, Technical*
- Working Conditions
 - Activity*
 - Independence*
 - Variety*
 - Compensation*
 - Security*
 - Working Conditions*

Using an Individual's Work Values to Identify Occupations to Explore

The purpose of the WIL and its corresponding occupational lists is to help clients explore occupations that are likely to reinforce their top work values. Each occupation in **O*NET OnLine** has been assigned ratings on how much its environment reinforces each work value and need (see ***Determining the***

Occupational Reinforcer Patterns for O*NET Occupational Units [McCloy, Waugh, Medsker, Wall, Rivkin, & Lewis, 1999a] for a detailed discussion of the development of this information). Providing clients with a listing of those occupations with environments most likely to reinforce their top work values allows clients to increase the likelihood of exploring careers and jobs that they are likely to find satisfying and rewarding.

What is the difference between work values and vocational interests?

Although some people have used the terms work values and vocational interests interchangeably, they **do not represent the same concepts**. Work values emphasize what is *important* or *unimportant* to an individual, whereas interests refer to what an individual *likes* or *dislikes*. Typically, questions used to assess values focus on *ends*, such as goals or standards, whereas questions used to assess interests focus on *means*, such as activities (Dawis, 1991). For example, a person who communicates an interest in doing volunteer work in a nursing home might express a value that it is important to be of service to other people.

Do individuals' abilities also have an important role in the theory?

Yes, in addition to needs/values, abilities serve a critical role in the Theory of Work Adjustment. The fit between an individual's abilities and the ability requirements of a job affects performance. In addition to taking the **O*NET Work Importance Locator**, clients who are exploring careers are encouraged to take an assessment tool, such as the **O*NET Ability Profiler**, to help them learn about their abilities (for more information on the **O*NET Ability Profiler**, contact the National Center for O*NET Development).

Note: Refer to the reference section of this guide for a listing of additional reading materials related to the Theory of Work Adjustment.

O*NET WORK IMPORTANCE LOCATOR SCORE REPORT

Below, each section of the **O*NET Work Importance Locator Score Report** is described.

Name and Date

Clients should write their name and the date in the spaces provided on page 1 immediately after they receive their **O*NET Work Importance Locator Score Report**. Recording this information will help clients to keep track of their Score Report.

The O*NET Work Importance Locator: How can it help you?

This section reminds clients that they completed the WIL to identify their work values. It tells them that the WIL had them rank different aspects of work that represent six important work values. The section informs clients that they are going to use their “highest” work values, which they identified and learned about at the end of the WIL, to explore the world of work.

Finally, this section reinforces the notion of the proper use of **O*NET Work Importance Locator** results. It reminds clients that results should be used for career exploration purposes only, and that they should not be used for employment selection or applicant screening.

What are your highest work value scores?

This section asks clients to write down the score and name of their two highest work values from page 6 of their WIL (Step 4). It also asks clients to transfer the same information to page 10 of the Score Report. Page 10 is a summary sheet designed to help clients gather all of the information they will need to explore careers.

Possible Client Challenges

- Clients might forget to write their work values information on the Summary Sheet (page 10).
- Some clients might have trouble copying their scores from the WIL.
 - ☞ Point out that the information they need to copy to their Score Report should be written in the yellow box on page 6 of the WIL.

Possible Client Challenges (continued)

- Clients' top work values may have the same score or have scores within 5 points of each other.



If clients' top work values have scores that are equal or within 5 points of each other, let clients know that they should start exploring careers with the work value they think best represents them based on their experiences.

- Clients might think the work values with their highest scores don't represent them.



Inform clients to keep working through the Score Report because, in later sections of the Score Report, options for changing or re-evaluating their WIL results are presented.

Using Your Work Values to Explore Careers

This section of the Score Report explains to clients that work values are useful when exploring careers because occupations that meet their work values are more likely to be satisfying and rewarding, as compared to occupations that do not. The section reminds clients that, in addition to work values, there is a variety of information about themselves that is useful when exploring careers. Clients are introduced to the concept of *whole-person assessment*—the more you know about yourself, the more likely it is that you can find satisfying work.

What Your Work Values Mean

This section of the Score Report provides clients with the summaries of the six work values, as listed below, and asks them to read, at the very least, the definitions for their two highest work values to get a better understanding of these work values for exploring jobs.

- **Achievement**—If Achievement is your highest work value, look for jobs that let you use your best abilities. Look for work where you can see the results of your efforts. Explore jobs where you can get the feeling of accomplishment.
- **Independence**—If Independence is your highest work value, look for jobs where you can do things on your own initiative. Explore work where you can make decisions on your own.
- **Recognition**—If Recognition is your highest work value, explore jobs with good possibilities for advancement. Look for work with prestige or with potential for leadership.
- **Relationships**—If Relationships is your highest work value, look for jobs where your co-workers are friendly. Look for work that lets

you be of service to others. Explore jobs that do not make you do anything that goes against your sense of right and wrong.

- **Support**—If Support is your highest work value, look for jobs where the company stands behind its workers and where the workers are comfortable with management's style of supervision. Explore work in companies with a reputation for competent, considerate, and fair management.
- **Working Conditions**—If Working Conditions is your highest work value, consider pay, job security, and good working conditions when looking at jobs. Look for work that suits your work style. Some people like to be busy all the time, or work alone, or have many different things to do. Explore jobs where you can take advantage of your particular work style.

What occupations are linked with your work values?

This is a critical section of the Score Report. It introduces clients to the process of linking their work values to occupations. First, it explains to clients that the Score Report contains six occupational lists, one for each work value. The occupations included in each list are those that will most likely reinforce or satisfy individuals with that particular work value.

Additionally, the concept of Job Zones is introduced to clients. Each of the six occupational lists are further divided into five Job Zones. Each Job Zone contains occupations that require similar levels of education, training, and experience. This information is important for clients to consider when exploring careers. It helps clients get an accurate picture of how much preparation is required to pursue certain occupations. Thus, when exploring careers, it is not enough just to have an occupation match your work values. You must also consider the amount of education, training, and experience needed to qualify for and be successful in occupations.

Note: Occupations were placed in Job Zones based on Specific Vocational Preparation (SVP) ratings located in the *Dictionary of Occupational Titles* (U.S. Department of Labor, 1991). For a further explanation of this process, see Oswald, McCloy, Lewis, and Rivkin (1999).

Note: In the Score Report and in the **Work Importance Locator O*NET Occupations Master List**, occupations are sorted under the different work values based on their occupational reinforcer pattern (for a further explanation of the occupational reinforcer patterns, see McCloy et al., 1999a). The majority of occupations are listed based on their Primary Work Value within Job Zones. To provide clients with a variety of occupations to explore (i.e., present 20 occupations per Work Value/Job Zone cell), where feasible, some occupations are included based on their Secondary or Tertiary Work Value. To make the Score Report less cumbersome for clients, a maximum of 20 occupations per Work Value/Job Zone cell are presented. For those cells for which more than 20 occupations were

present, a sample of occupations was drawn. Clients interested in the entire listing of occupations should refer to the **Work Importance Locator O*NET Occupations Master List**.

What is a Job Zone?

In this section clients learn more about why occupations are placed within Job Zones and how Job Zones can help them focus their career search. Clients are given a brief definition of each Job Zone and are introduced to the concepts of Current Job Zone and Future Job Zone.

Clients' Current Job Zones are comparable to the amount of education, training, and experience they have now. They can consider this amount of preparation when exploring careers, or they can use their Future Job Zone—the amount of education, training, and experience they expect to have in the future, after they finish high school, college, or a vocational training program.

For clients with less work experience, use of their Future Job Zone for exploring careers is probably more appropriate. This will give them broader exploration opportunities and will more accurately reflect their career aspirations. However, the Score Report does *not* tell clients which Job Zone type they should use to explore careers. It leaves the choice up to them.

Possible Client Challenges

- Clients might have a hard time deciding whether to use their Current or Future Job Zones to explore careers.
 - ☞ Suggest that clients with little work experience, who have not yet decided how much education to pursue, select a Future Job Zone. This will allow them to open their minds to possible careers that they may not have considered. Clients will get a better understanding of how “higher” job zones affect career choices.
- Clients may have a difficult time understanding the concept of Future Job Zone.
 - ☞ Encourage clients to think about themselves in the future. What do they want their lives to be like as adults? Encourage them to think about the “most” they can do in their lives.

Job Zone Definitions

This section of the Score Report contains complete definitions of the Job Zones. By reading each definition, clients will begin to learn the differences among the five Job Zones. As they move from Job Zone 1 to Job Zone 5, more experience, education, and training are required. For example, many occupations in Job Zone 1 require either a GED or a high school diploma, frequently need very little previous work-related experience, and usually involve simple training that can be delivered by a co-worker. Contrastingly, Job Zone 5 includes occupations that need the most overall preparation. These occupations frequently require advanced degrees, such as Ph.D., M.D., or J.D., and at least four years of work-related experience.

Possible Client Challenges

- Clients might have difficulty distinguishing between Job Zones that are next to each other (e.g., the difference between Job Zones 3 and 4).
 - ☞ Clients should pay particular attention to the “Overall Experience” category within Job Zones. This section of the definition can help clients figure out the differences among Job Zones.
- Clients might need more “familiar” examples of occupations that fit into a particular Job Zone to really understand the Zone.
 - ☞ You may want to look at occupational “snapshots,” which are included in **O*NET OnLine**, to find other examples of occupations within a Job Zone.

Which Job Zone suits you best?

This section has clients select the Job Zone that they want to use to explore careers. Clients are asked several questions to help them focus on a Job Zone. Once again, they are instructed to think about whether they want to use their Current Job Zone or Future Job Zone to explore careers.

Clients are instructed to write down the Job Zone(s) they select in the spaces provided on the summary sheet located on page 10 of the Score Report.

Possible Client Challenges

- Clients might have difficulty understanding the differences between Job Zones.
 - ☞ As discussed in the Job Zone Definition section above, clients might need more examples of occupations within a Job Zone in order to really understand the Zones.
- Clients might have difficulty deciding whether they should use their Current or Future Job Zone to explore careers.
 - ☞ Clients should be encouraged to use a Future Job Zone because this will provide broader career exploration opportunities. For some clients, it might be helpful to explore a variety of Job Zones, so they can see how their career opportunities expand as they achieve more education, training, and experience.

Exploring Careers Using Your Work Values and Your Job Zone

Now clients are ready to use their work values and Job Zone(s) to explore careers. This section of the Score Report provides clients with a step-by-step description for using the information they discovered about themselves to explore occupations. These steps are summarized below.

1. Look at your highest work value occupations.

Clients are informed that the occupational lists begin on page 11. They then are directed to look through the six occupational lists until they find the list with the title that corresponds with their highest work value. The section lets them know that these occupations have the best chance of satisfying their work values.

2. Review the occupations in your Job Zone.

Clients are directed to find the section of their work values list that matches their Job Zone. They are reminded that a Job Zone will help them find occupations that will make the best use of either their current or future knowledge and skills. Clients are encouraged to use the Summary Sheet on page 10 of the Score Report to write down the titles of occupations that they want to explore and find more information about.

Note: Each list of occupations is clearly divided into the five Job Zones. Clients should be able to find their Job Zone section easily.

3. Find out more about the occupations.

Here clients are instructed to look in **O*NET OnLine** to find out information about the occupations they want to explore. Information about occupations that clients can learn about includes: work activities, skills, and the wage/future employment outlook.

Note: Clients should be able to use **O*NET OnLine** on their own for career exploration. You may, however, wish to hold an **O*NET OnLine** training session to help them locate the most important information in **O*NET OnLine** for career exploration. For more information on **O*NET OnLine**, contact the National Center for O*NET Development (contact information listed on page 2 of this Guide).

4. Check the Work Importance Locator O*NET Occupations Master List.

This list is configured just like the lists in the Score Report, but it contains more occupations. Clients can look at the Master List to find more occupations linked to their work values and Job Zones. Clients are told that they can get this list from their teacher or counselor.

5. Still want more? Check your Job Zone and your next highest work value.

Like Step 4, this step gives clients more options to find other occupations that they might like to explore. They are told to review their Job Zone selection to see if another Job Zone might be more appropriate. They are also told to explore occupations under their next highest work value. Finally, they are reminded to refer to the Master List for more occupations.

Note: The purpose of Step 5 is to encourage clients to expand their career exploration and to give them options for exploring occupations that they might find satisfying and rewarding.

6. Is there a particular occupation you want to explore?

Some clients may have certain occupations that they want to explore which do not appear in their occupational lists because the occupations do not match their work values or Job Zone. Clients are directed to find the list of occupations where the occupation they want to explore is placed. They are reminded that, within each list, the occupations are presented in alphabetical order. After clients find the occupation, they will know which work value it is most likely to satisfy. The section directs clients to go back to page 5 of their WIL and look up their score for that work value. Clients are informed that if the score is less than 15, they are less likely to find this particular occupation satisfying.

7. Want to know about a particular occupation not on your list?

Clients also can use **O*NET OnLine** to learn more about an occupation not on their list. This section of the Score Report informs clients that they can find those occupations in the **Snapshot** part of **O*NET OnLine**. The section also contains a series of questions that clients may want to consider when looking at a particular occupation. For example, how high is their score for the occupation's work value? Is the score for that work value close to their two highest scores? Clients are informed that if the work value is one of their four lowest scores on the WIL, it may not be suited to their work values, and that they may want to think about why they selected this occupation to explore.

The section also directs clients to consider what they think is important about this particular job. Examples presented include: money, glamour, excitement. Clients are encouraged to explore the occupation further to ensure that the work will be satisfying to them. Clients are provided with suggestions for learning more about the occupation, including reading about the occupation in other reference materials, talking to someone who is actually in the job, or taking additional **O*NET Career Exploration Tools** to see how different pieces of information about themselves match the occupation. The options are presented to clients to facilitate their ability to better assess a particular occupation. By following these options, clients can develop more viable career aspirations.

Not really sure you agree with your results?

This is an optional section for clients. Clients who have confidence in their results are directed to skip forward to page 9 of the Score Report.

Some clients, however, may decide that their WIL results do not accurately reflect their work values. This section gives clients options designed to help them become more comfortable with their WIL results. The goal of this section is to make sure that clients continue to explore careers even if they are not happy with their WIL results. The WIL was developed to encourage, not discourage, clients' career aspirations.

The options that clients can follow are presented below:

1. Check your card sorting.

Clients are directed to reread the instructions for Step 2 of their WIL and to verify that they sorted their cards correctly. Clients who still have the cards sorted onto their **Work Value Card Sorting Sheet** are asked the following questions to help them make sure they completed the card sorting properly:

- Did they put exactly four cards into each column?
- Are the most important work value cards in Column 5?
- Are the next most important work value cards in Column 4?
- Are the next most important work value cards in Columns 3 and 2?
- Are the least important work value cards in Column 1?

2. Check your scoring.

Clients are directed to reread the instructions for Step 3 and to verify that they scored their WIL correctly. Clients are encouraged to double-check their answers to make sure that they really understood the instructions. They are informed that their scores should fall between 6 and 30. Clients then are asked the following questions to help them make sure that they scored the instrument properly:

- Did they put the correct column number (5, 4, 3, 2, or 1) next to the matching work value card letter on their Work Value Worksheet?

- Did they add up the scores within each colored box correctly? Clients are encouraged to check their addition.
- Did they multiply the sum of their Work Values Scores correctly? Clients are encouraged to check their multiplication.
- Did they multiply the Working Conditions value by mistake? Clients are reminded the total score for Working Conditions does **not** need to be multiplied.
- Did they copy their highest Work Value Scores from Step 3 to Step 4 (page 6 of the WIL) correctly?

3. Take another look at the occupations listed for your highest work value.

This section encourages clients to take another look at the occupations listed for their highest work value. By taking a closer look at the occupations on the list, clients may find that there are, in fact, some occupations with their highest work value that are worth exploring.

Clients are asked to consider if they are interested in any of the occupations listed. Are the clients familiar with all of the occupations on the list? Clients are reminded that **O*NET OnLine** can provide them with additional information.

4. Use your next highest work value to explore careers.

Clients are presented the option to use their next highest work value to explore careers. They are asked to read the definition of their next highest work value, and think if it describes their values better than their highest work value. Clients are also directed to look at the occupations that are listed under their next highest work value. The section asks clients to consider whether the occupations appear to be “more in line” with their values, and whether there are any occupations they would want to explore further.

5. Try your work value results out.

This section encourages clients not to “give up” on their results too soon. It points out that once clients use the results of their WIL to explore careers, they may find that their scores make sense.

Clients are encouraged to:

- Explore some of the occupations linked to their highest work value to determine what the jobs really entail and to see if the occupations have any of the characteristics that the clients find important in work.
- Talk to some people who work in one of the occupations linked with their highest work value. This can give them a better idea about what is actually involved in performing the occupation.

Using Your Work Values with Other Career Exploration Tools

This section introduces the idea of using WIL results along with **O*NET Interest Profiler** (IP) results. Clients are presented with the notion of the **O*NET Occupations Combined List: Interests and Work Values**, which allows them to use both their WIL and IP results to explore careers. A list of other career exploration tools is provided to clients, and the idea of using different pieces of information about themselves to explore careers is reinforced.

O*NET Occupations

In this final section of the Score Report, clients are presented six work values occupational lists, each divided into five Job Zones. Occupations within a Job Zone section are linked to that Job Zone, as well as to the overall work values area the list represents. Each occupation is described by an O*NET title and occupational code. Clients can use either the title or the code to find the occupation in **O*NET OnLine**.

On the Summary Sheet located on page 10, clients are instructed to record (if they haven't already done so) their highest work values and their current and future Job Zones. They are also instructed to record in the space provided the occupations that they have selected to explore.

In general, the Job Zones within each work value occupational list contain approximately 15 to 20 O*NET occupations. These occupations were selected because they are representative of the total group of occupations that are listed in the corresponding section of the **Work Importance Locator O*NET Occupations Master List**. They also were selected because they represent different areas of the world-of-work.

The majority of occupations were assigned to a particular list because their work environment was rated highest for that value. However, some occupations also were assigned to a list based on their environment's second and/or third value. These occupations are identified by either a double or triple asterisk.

NOTE: Due to the assignment procedure, an occupation can be located in multiple work value lists.

SUGGESTIONS FOR SCORE INTERPRETATION PROGRAM ACTIVITIES

The WIL Score Report also can be used to facilitate group sessions aimed at helping your clients interpret their results. For example, you can work step-by-step with clients through the Score Report. Clients can take turns reading sections or discussing their feelings about their results. Another option is to work with your clients to help them decide whether they want to explore additional occupations using other Job Zones or other work values.

To help clients better understand their work value results, activities that you can incorporate into your career exploration/vocational training program are presented below. These activities can be used in a group setting or individually by clients, or you might decide to have clients work on some sections of the activities independently and on other parts in a group.

- **Use different Job Zones and work values to explore careers.**

Clients can try out other Job Zones and work values to explore careers. This can help them confirm their results (e.g., they can see that the other work values/Job Zones do not contain occupations that they wish to explore), or they might find other work values/Job Zones that they want to use to explore occupations. Finally, this activity can give them a broader picture of the world-of-work.

- **Clients can try their results out.**

Clients could visit an individual working in one of the occupations they are thinking of pursuing. They could actually see what the person does on the job. They can talk to the employees in the job to learn about the work values that the employees have and how they compare to their own work values. You might want to have clients prepare a list of questions to ask employees.

If clients can't visit a person in the job, they could find a friend or relative in a job they may wish to pursue. They could talk to the person and report back to the class:

- 1) what the person thinks is important about the job,
- 2) what the person does on the job, and
- 3) what the person's work values and interests seem to be.

- **Use America's Job Bank (AJB), and other sources to look for jobs.**

AJB, which can be found on the Internet, <http://www.ajb.org>, helps clients to see the types of job openings that exist for occupations they have selected to explore. If clients do not have access to the Internet through their school or home, they can go to their local employment service office to gain access to AJB. Clients also can try to find their occupations in the classified section of a newspaper.

- **Identify the highest work value of an occupation using the six work values.**

Provide clients with a list of occupations. Have each client assign a top work value to each occupation. In a group, have clients discuss the characteristics about each job that caused them to assign their rating.

- **Use O*NET OnLine to explore occupations.**

Clients can explore particular occupations using **O*NET OnLine**, <http://www.onetcenter.org>. They can be directed to find other information about the occupation that supports their view that the occupation is indeed one that they should pursue further. They can also be instructed to find information that supports their view that a particular occupation is not right for them.

- **Use America's Learning eXchange (ALX) to find courses related to occupations.**
ALX, which can be found on the Internet, <http://www.alx.org> , allows users to see training programs and courses available for the occupations they have chosen to explore. They also can learn about requirements for licensing or certification.

WHERE CAN I GET MORE HELP WITH USING THE O*NET WORK IMPORTANCE LOCATOR?

For more information about the **O*NET Work Importance Locator** and its Score Report, contact the Center for O*NET Development:

Internet:
<http://www.onetcenter.org>

Mail:
Customer Service
National Center for O*NET Development
Post Office Box 27625
Raleigh, NC 27611

E-mail:
o*net@esc.state.nc.us

FAX: (919) 715-0778

DEVELOPMENT OF THE O*NET WORK IMPORTANCE LOCATOR

Two primary goals served as the basis for the development process of the **O*NET Work Importance Locator**:

- 1) Create a self-scoring, self assessment instrument that measures the work values identified by the Theory of Work Adjustment (Dawis & Lofquist, 1984). The WIL is based on a previously developed measure of work values, the Minnesota Importance Questionnaire (MIQ; Rounds, Henly, Dawis, Lofquist, & Weiss, 1981). However, the MIQ is not well suited for self- assessment. In particular, its scoring system is complex and, therefore, can be performed only by the publisher. Steps were taken to design the WIL in a manner that allows clients to score their work values immediately after its completion and to ensure that clients can understand how to complete and interpret the results of the instrument without the aid of a counselor or teacher.
- 2) Develop an instrument that reliably and accurately measures the work values identified by the Theory of Work Adjustment (Dawis & Lofquist, 1984). While the WIL is based on the MIQ, modifications to the items and procedures were necessary. Steps were taken to ensure and verify that such alterations did not compromise the instrument's psychometric characteristics, including the test-retest reliability, internal consistency, and accuracy of its scores.

The following sections summarize the development and evaluation of the WIL. For a more technical and comprehensive description of the WIL, refer to following technical report available from the National Center for O*NET Development: ***Development of the O*NET Work Importance Locator*** (McCloy et al., 1999b).

DEVELOPMENT

The WIL uses a card-sorting task that is similar to the Q-sort technique (Stephenson, 1953). Clients complete the WIL by sorting 20 cards, each containing a description of a need statement, in terms of their relative importance in the client's ideal job. The following are examples of need statements:

- "My pay would compare well with that of other workers."
- "I could do things for other people."
- "I could be busy all the time."
- "I could try out my ideas."

- “The job would provide an opportunity for advancement.”

Before reaching its current form, several iterations of the WIL were developed, modified, and enhanced based on customer feedback from multiple pilot studies. These modification stages are briefly described below.

Initial Version

As stated previously, the WIL was based on the Minnesota Importance Questionnaire (MIQ; Rounds, Henly, Dawis, Lofquist, & Weiss, 1981). All items or need statements contained in the WIL were based on the original 21 need statements from the MIQ. Many of the items, however, were worded somewhat differently from their MIQ source items. For the most part, the items were modified to match statements describing work values that are found in **O*NET OnLine**. Table 1 provides the new wording of the need statements, along with the original wording of each statement from the MIQ.

Table 1.
Wording Changes for the 21 Need Statements

WIL	Original MIQ Items
1. On my ideal job it is important that I <u>make use of my abilities</u> . ¹	1. On my ideal job it is important that I <u>could do something that makes use of my abilities</u> .
2. On my ideal job it is important that the <u>work</u> could give me a feeling of accomplishment. ¹	2. On my ideal job it is important that the <u>job</u> would give me a feeling of accomplishment.
3. On my ideal job it is important that I could be busy all the time.	3. On my ideal job it is important that I could be busy all the time.
4. On my ideal job it is important that the job would provide an opportunity for advancement.	4. On my ideal job it is important that the job would provide an opportunity for advancement.
5. On my ideal job it is important that I <u>could give directions to others</u> . ²	5. On my ideal job it is important that I <u>could tell people what to do</u> .
6. On my ideal job it is important that I <u>would be treated fairly by the company</u> . ²	6. On my ideal job it is important that <u>the company would administer its policies fairly</u> .
7. On my ideal job it is important that my pay would compare well with that of other workers.	7. On my ideal job it is important that my pay would compare well with that of other workers.
8. On my ideal job it is important that my co-workers would be <u>easy to get along with</u> . ²	8. On my ideal job it is important that my co-workers would be <u>easy to make friends with</u> .

WIL	Original MIQ Items
9. On my ideal job it is important that I could <u>try</u> out my own ideas. ¹	9. On my ideal job it is important that I could <u>try out some of my own ideas</u> .
10. On my ideal job it is important that I could <u>work alone</u> . ¹	10. On my ideal job it is important that I could <u>work alone on the job</u> .
11. On my ideal job it is important that I would never <u>be pressured to do things that go against my sense of right and wrong</u> . ³	11. On my ideal job it is important that I <u>could do the work without feeling that it is morally wrong</u> .
12. On my ideal job it is important that I could <u>receive</u> recognition for the work I do. ¹	12. On my ideal job it is important that I could <u>get</u> recognition for the work I do.
13. On my ideal job it is important that I could make decisions on my own.	13. On my ideal job it is important that I could make decisions on my own.
14. On my ideal job it is important that the job would provide for steady employment.	14. On my ideal job it is important that the job would provide for steady employment.
15. On my ideal job it is important that I could do things for other people.	15. On my ideal job it is important that I could do things for other people.
16. On my ideal job it is important that I <u>would be looked up to by others in my company and my community</u> . ²	16. On my ideal job it is important that I <u>could be "somebody" in the community</u> .
17. On my ideal job it is important that I <u>have supervisors who would back up their workers with management</u> .	17. On my ideal job it is important that <u>my boss would back up the workers (with top management)</u> .
18. On my ideal job it is important that I <u>would have supervisors who train workers well</u> . ¹	18. On my ideal job it is important that <u>my boss would train their workers well</u> .
19. On my ideal job it is important that I could do something different every day.	19. On my ideal job it is important that I could do something different every day.
20. On my ideal job it is important that the job would have good working conditions.	20. On my ideal job it is important that the job would have good working conditions.
21. On my ideal job it is important that I could plan my work with little supervision.	21. On my ideal job it is important that I could plan my work with little supervision.

¹ Minor difference in the wording between WIL and MIQ

² Moderate difference in the wording between WIL and MIQ

³ Substantial difference in wording between WIL and MIQ

Some rewording also was necessary for the associated value labels (to ensure that these labels were readily interpretable by all clients). During the initial WIL development efforts, the value labels used herein (achievement, working conditions, recognition, relationships, support, and independence) had been referred to by their more traditional psychological construct labels (respectively: achievement, comfort, status, altruism, safety, and autonomy). Also, in the initial form of the WIL that was developed, clients sorted all 21 needs into seven card piles of three cards each. This design preserved all 21 of the MIQ needs, but required more complex scoring. To compute the score for each value, the client had to refer to a table (which

showed the weighted score for each need) and then add up three two-digit numbers. The table made it unnecessary for the respondent to do any multiplication. The numbers in the table took into account the pile number and the number of items in the value scale.

However, after reviewing this initial form, it was thought that the seven-pile scoring procedure would be too complicated for some clients because they would have to look up numbers in a table and add two-digit numbers.

Five-column Version

A second version of the measure was developed that omitted the table lookup and required the addition of only one-digit numbers. This version was very similar to the final form of the WIL. The two possible disadvantages of this design were: a) one of the items (need statements) would have to be dropped in order to keep the math simple, and b) it required multiplication.

After a discussion with one of the MIQ's co-authors, it was concluded that the loss of one carefully chosen item (Item 16, the need for social status) would have very little effect on the quality of the measure. Item 16 was chosen because: a) the wording had changed at least moderately from the original MIQ wording, and b) its deletion would not yield another two-item value (i.e., it was from a scale with at least four items). With the rewording, Item 16 appeared to have a different meaning from other items meant to measure social status (see Table 1). With regard to the computation required, the largest multiplications that might be required with a 20 item version were 2×15 and 3×10 .

Based on the points stated above, materials for a five-column version of the WIL were developed. The materials consisted of: a) two pages of instructions; b) 20 cards, each of which displayed a need statement and had a letter A through T printed on it; c) a card sorter sheet used to place the cards into five groups; and d) a scoring page on which need scores were reported and value scores were calculated.

When a client sorts the 20 cards of need statements into five columns (four cards per column), the four needs that are the most important are placed in the first column, the four needs that are next in importance are placed in the second column, and so on. The client then records the column number for each card on the scoring sheet. The column number represents the score for that need (e.g., each need in the *most important* column gets a score of 5). There are six tables on the scoring sheet. The scoring sheet is laid out so that cards that represent the same work value are grouped together in the same table. After all the column numbers have been recorded, the respondent computes the six value scores by adding one-digit numbers and multiplying the one or two-digit sums.

Pre-pilot Study

A pre-pilot study of the WIL was conducted to determine how easy the WIL was to use (e.g., were the instructions easy to follow?) and to identify possible improvements to the design and procedure of the instrument. In addition, information was gathered on the participants' reactions to the WIL.

The measure was administered to 21 employment service clients in North Carolina. The information on the participant's scoring sheet was checked for errors, including the number of cards placed in a column, as well as errors occurring in the addition and multiplication steps of the scoring procedure. Nine people made at least one error while taking the measure. The number of people making each type of error is shown in Table 2 below.

Table 2.

Number of People Making Errors in the WIL Pre-Pilot Study

Type of Error	Number of People Who Made an Error
Addition	4
Multiplication	2
Wrong number of cards in a pile	5
Multiplied each pile number by four before writing it on the scoring sheet	1
Any type of error	9

Note. $N = 21$. Three people made two types of errors.

Participant errors discovered in the pre-pilot led to modifications of the WIL. The large number participants placing the wrong number of cards in a pile was a major concern. The participants were supposed to put four cards into each of the five rectangles printed on the Work Value Card Sorting Sheet. Some people put the wrong number of cards in some rectangles. In order to reduce the likelihood of this type of error, the Work Value Card Sorting Sheet was modified so that there were 20 rectangles printed on the sheet. Clients would then be directed to place each card into its own rectangle, making the placement of four cards per column more obvious. The math errors were addressed by improving the instructions of the WIL and making minor format changes to the scoring sheet to make it easier for clients to understand which numbers to add and which numbers to multiply.

The pre-pilot participants also completed a Participant Reaction Questionnaire. Only one participant found the task boring or tiring; two became frustrated with the task; and one was frustrated that some important work values were not explicitly included in the measure (e.g., flexible scheduling and leave policy).

Almost all of the respondents said the instructions were either *clear* or *very clear*. Only two of the six people who made column or score transfer errors thought the instructions were *very clear*, suggesting that an improvement in the instructions would probably reduce errors.

Pilot Study

After the WIL had been modified, it was tested in the pilot study at an employment service center in Utah. The WIL and a Participant Reaction Questionnaire were completed by 48 clients of the center.

The test administrators made the following observations during WIL pilot testing:

- Most participants completed the task without difficulty.
- Mathematical errors in score computation were the most common problem.
- Some participants expressed frustration with having to rank some cards on the lower end of the Importance Scale.
- Average time to complete = 14 minutes.

Errors made by the participants of the pilot study are detailed in Table 3. The Participant Reaction Questionnaire results for the WIL are shown in Table 4.

As Table 3 shows, the number of apparent errors decreased from the level in the pre-pilot study, with sorting errors disappearing entirely. Unfortunately, the test administrators stated in a debriefing that they had helped many of the respondents. Therefore, the number of errors probably would have been higher if all of the respondents had completed the measure without assistance. The administrators speculated that the number of errors would have been as high as 30%; the actual error rate is not known.

Table 3.

Number of People Making Errors in the WIL Pilot Study

Type of Error	Number of People Who Made an Error
Addition	7
Multiplication	2
Wrong number of cards in a pile	0
Multiplied the total for Work Value 2 (This total should not be multiplied)	1
Any type of error	8

Note. $N = 48$. Two people made 2 types of errors.

Table 4.

Percentage of Participants Who Responded Positively to Pilot Study Reaction Questionnaire

	Percent Positive Responses
Evaluation Question	Paper Version (N = 48)
How clear were the instructions on the survey?	90%
Did you find the rankings easy to do?	92%
Was the survey easy to score?	95%
Did you get tired or bored at any time during the survey?	100%
Are the results of the survey consistent with how you would describe yourself?	87%

Final Version

Further revisions to the WIL were made based on the feedback from the participants and administrators of the pilot study. These changes were small, but affected several parts of the measure. The final version of the WIL consists of:

- 20 needs statement cards, labeled A through T;
- a 3-page booklet that contains the instructions and scoring page; and
- the Work Value Card Sorting Sheet (11-inch × 17-inch) that also includes some instructions.

Summary

In summary, the WIL was developed based on items from the MIQ, with one item dropped to make it easier for respondents to self-score their results. The pre-pilot and pilot studies were conducted to gain information to improve the measure before administering it to a larger sample to gather information on its psychometric characteristics.

EVALUATION

The evaluation of the WIL's psychometric characteristics was part of a large, multi-stage study that also served to gather data related to the **O*NET Work Importance Profiler (WIP)**, the computerized version of the WIL. The portions of this large scale study that are relevant to the evaluation of the WIL will be summarized in the following sections. For a detailed examination of the entire study's design, as well as for a more extensive presentation of results related to the WIL, refer to the report, ***Development of the O*NET Work Importance Locator*** (McCloy et al., 1999b).

Reliability

The term "reliability" refers to the degree to which a measurement procedure is free from unsystematic errors of measurement and the degree to which it gives the same values if the measurement procedure is repeated. An individual responding to a measure is likely to have different results if he or she took the instrument again. Systematic differences in scores (e.g., improvement on a test taken at two different times because the individual's knowledge has increased between tests) should not be considered the *unreliability* of a measure. But an individual's results may change when measured more than once on the same measure because of unsystematic effects (e.g., miss-marking a response to an item; feeling tired one day, but not the next). Such unsystematic differences are considered unreliability. Low reliability limits the ability to have confidence about individuals' results from a single measurement (i.e., results may or may not lack precision). The higher the reliability of a measure, the more confidence you can have in the information obtained from the measure.

There are several ways to assess the reliability of measurement, depending on the type of consistency with which one is most concerned, including *test-retest reliability*, *alternate or parallel forms reliability*, and *internal consistency*. The following subsections will present the evidence gathered on the WIL that related to each of these types of reliability.

Test-retest Reliability. This type of reliability refers to the consistency of results when the same individual is assessed on the same measure at two points in time. This information is obtained by looking at the degree of relationship (i.e., correlation) between an examinee's scores obtained on the measure at different points in time. Estimates of test-retest reliability are particularly useful if the characteristic being measured is not expected to change over the time between the two measurement periods (e.g., a measure of personality characteristics of normal adults at two points in time that are a month apart, as opposed to a measure of knowledge administered before and after a course on the subject of the measure). Given that work values of adults are considered to be relatively stable characteristics, it would be expected that individuals' responses to the WIL should be stable across time.

Two hundred and thirty vocational/technical and community college students were administered the WIL twice, with a two-month interval between the first administration and the second administration. Evidence of the WIL's ability to reliably measure individuals' top-ranked work value was moderately high, with a person's top work value being the same between administrations 62 percent of the time. However, the correlation for the first administration's six work value scores and the second administration's six work values scores ranged between .35 (Achievement) and .58 (Support), indicating that the WIL has a low-to-moderate ability to reliably measure each of the six work values over the two month interval. Overall, this evidence reinforced the use of the WIL to help clients discover their highest work value, while also demonstrating that the WIL should not be used by clients to determine the rank order or profile of all six of their work values.

Alternate or Parallel Forms Reliability. This type of reliability is the evaluation of similar responding by the same individuals on forms which have been created to be *alternative* or *parallel* forms of the same measure. This estimate of reliability was important because the WIL and the WIP (i.e., the O*NET computerized measure of work values) were designed to be used interchangeably, depending on the computer resources of the location where the measurement is taking place. Similar results for the same individuals on these different measures would support using the measures interchangeably.

The same sample of 230 vocational/technical and community college students used in the test-retest reliability study described above were also administered the computerized work value measure (WIP), allowing them to provide data relevant to alternate forms reliability. The scores of the two measures were reformulated in a manner that allowed for direct comparison and corrected for "ipsatization" problems (this correction reduces the adverse effects of forced-choice rank order information on a correlation coefficient (e.g., it reflects the impaired ability of clients to rate associated needs in similar ways given that they have used up the available spaces at their preferred level of importance). The six work value scores derived from both measures had correlations ranging from .70 to .80, with a median correlation of .77. This indicates a relatively high agreement for the measurement of values between the two work values measures.

Internal Consistency. This type of reliability is used to determine whether different items, which are measuring the same subject on the same measure, have highly related results. For example, if a test included 10 items on addition and 10 items on reading ability, one would expect to see higher interrelationships within the set of 10 addition items and within the set of 10 reading ability items than between items from the two different sets. Thus, internal consistency reliability is another type of reliability analysis which can be applied to the WIL to assess the adequacy of its development. In terms of the WIL, it would be desirable to have high internal consistencies among items within the same scale (i.e., the needs that are used to measure each of the six work values).

The responses of 1,199 employment service clients and junior college students drawn from 23 sites were used to examine the internal consistency of the WIL. While the examination of internal consistency is important, the rank order format of the WIL provides data that, for statistical reasons, inhibit its ability to demonstrate high internal consistency values. The rank order format leads to the presence of negative inter-item correlations, attenuating the measurement of internal consistency reliability. The median coefficient alpha obtained for the sample was .20, indicating a very low level of internal consistency. An examination of coefficient alphas for each of the six scales after the data were "corrected for ipsatization" (i.e., reducing the adverse effects of rank order information) yielded an average increase of .38 per scale, indicating that while the rank order format did in fact adversely affect the coefficient alphas, the internal consistency of the six scales was, at best, moderate.

Summary. Overall, the WIL demonstrated moderate reliability across the majority of reliability analyses. The test-retest results showed moderate correspondence within individuals administered the WIL at two times several weeks apart. Individuals had the same top value 62 percent of time. After the effects of ipsatization were adjusted for, the correlations between the WIL and the WIP (computerized version) were in the .70's and .80's, with a median of .77 (indicating that the measures do have a degree of interchangeability). Internal consistencies were low, with a median value of .20, due in part to the effects of ipsatization.

Preliminary Validity Evidence

The term validity, *as used in this guide*, refers to whether or not an instrument functions as it was intended to function. The WIL was designed to measure work values in the same way as the MIQ and as defined by the Theory of Work Adjustment (Dawis & Lofquist, 1984). Validity analyses for the WIL focused on determining the degree to which the WIL and the MIQ appeared to be measuring the same work values constructs. Further evidence of validity should be forthcoming in the next few years.

Data from a sample of 550 employment service unemployment insurance clients, displaced workers, vocational/technical students, and community college students were used to analyze the similarity of scores obtained by the WIL and the MIQ. Correlations between the scores obtained by both instruments for the work values ranged from .30 to .49, which is fairly low. Two potential explanations for the low correlations exist: 1) the previously mentioned attenuative impact of the WIL's rank order format on the correlation coefficient, and 2) the effect of the wording modifications that were made to the need statement items of the WIL. An examination of the correlations at the needs level indicated that, with one exception, all items with a correlation of .40 or less contained modified text. This indicates that, at least to a small extent, the wording changes adversely affected relations between the WIL and MIQ.

In order to examine the validity of the WIL (as defined above) without the impact of its rank format, the consistency of the top work values obtained by the WIL and MIQ were compared. The top value was the same for each pair of instruments approximately 57 percent of the time. The top value obtained on the WIL was one of the top two values indicated by the MIQ 79 percent of the time. However, the top two values on one of the measures matched the top *two* values on the other measure 16 percent of the time.

Summary. The ability for the WIL to provide clients with a valid indication of their highest work value was supported. However, similar to the conclusions drawn after the examination of the WIL's reliability, validation evidence did not support clients' use of their results to determine the entire profile of their work values. While more direct validity evidence should be forthcoming, at present the user is cautioned to keep in mind the moderate relations just described. That said, given the demonstrated relations between the WIL, the WIP, and the MIQ, the interested reader may benefit from reviewing MIQ validity evidence in Rounds et al. (1981).

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